

Sodium

Test Details

METHODOLOGY

Ion-selective electrode (ISE); flame photometer

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Electrolyte, acid-base balance; water balance; water intoxication; diagnose dehydration.

Hypernatremia occurs in dehydration. For instance, nasogastric protein feeding with insufficient fluids may cause hypernatremia. Hypernatremia without obvious cause may relate to Cushing syndrome, central or nephrogenic diabetes insipidus with insufficient fluids, primary aldosteronism, and other diseases. Severe hypernatremia may be associated with volume contraction, lactic acidosis, azotemia, weight loss, and increased hematocrit as evidence of dehydration. The corrected serum sodium is often high in nonketotic hyperosmolar coma. (A corrected Na^+ is calculated by increasing Na^+ by 1.3–1.6 mmol/L for each 100 mg/dL increment in serum or plasma glucose). 100 mg equals 5.56 mmol/L. The corrected serum sodium level calculated in nonketotic hyperosmolar coma: apparent mild hyponatremia with very high glucose may actually mean (corrected) hypernatremia.¹

Hyponatremia occurs with nephrotic syndrome, cachexia, hypoproteinemia, intravenous glucose infusion, in congestive heart failure, and other clinical entities. Serum sodium is a predictor of cardiovascular mortality in patients in severe congestive heart failure.²

Hyponatremia without congestive failure or dehydration may occur with hypothyroidism, the syndrome of inappropriate secretion of antidiuretic hormone (SIADH), renal failure, or renal sodium loss.

The differential diagnosis of hyponatremia includes Addison disease, hypopituitarism, liver disease including cirrhosis, hypertriglyceridemia, and psychogenic polydipsia. Diuretics and other drugs may cause hyponatremia. Sodium decreasing to levels <115 mmol/L can lead to significant neurological dysfunction with cerebral edema and increased intracranial pressure.

The differential diagnosis of hyponatremia includes determination of urine sodium and osmolality and serum urea nitrogen (BUN). BUN is often decreased in SIADH.

Custom Additional Information

The ratio of serum sodium:osmolality is normally 0.43–0.50; a decreased ratio is found in uremia and other states in which there are increased substances with osmotic activity.

See Urea Nitrogen [001040] (/tests/001040/urea-nitrogen), regarding hyponatremia with sodium <128 mmol/L, hypo-osmolality, low BUN and the syndrome of inappropriate secretion of antidiuretic hormone.

A number of situations result in “pseudohyponatremia.” In these circumstances treatment may be undesirable. With pseudohyponatremia serum sodium is decreased but the serum is not hypotonic (serum osmolality is normal or even increased). This may occur as the result of other molecules replacing water in relation to sodium. The water content is effectively lowered – sodium is “diluted.” In severe hypertriglyceridemia or paraprotein-related marked increase in protein, the concentration of sodium in relation to water is normal but the analytic result is determined as mmol/L of serum. Osmolality in this situation is determined as amount of particles per kg of water and will be normal. It has been shown that analyses by sodium electrode of the direct potentiometric type (requires no dilution) are not artifactually low in patients with hyperlipidemia.³ If large amounts of solute such as glucose or mannitol are present, movement of intracellular water into the extracellular space may produce dilutional hyponatremia. In this case sodium concentration in relation to water is actually low. “Osmolal gap” however exists between measured and calculated serum osmolality. Other substances capable of increasing serum osmolality (eg, ethanol) may also cause increase in the osmolal gap. Yet another cause of pseudohyponatremia is increased serum viscosity due to increased globulin proteins, occurring particularly in Waldenström macroglobulinemia. The sodium analyzer may aspirate too little sample when viscosity is so increased, leading to a factitious low sodium concentration. See discussion of “pseudohyponatremia” by Epstein and Osler.⁴

Hyponatremia may manifest lethal neurological complications (water intoxication with brain edema). Rapid correction of hyponatremia has been described⁵ but has also been implicated as a cause of demyelination.⁶

Hypernatremia may complicate some cases of lactulose-treated portal-systemic encephalopathy.⁷

Drug effects are summarized.⁸

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection. Label specimen as serum.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

134–144 mmol/L

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Gross hemolysis; improper labeling

Footnotes

1. Daugirdas JT, Kronfol NO, Tzamaloukas AH, Ing TS. Hyperosmolar coma: Cellular dehydration and the serum sodium concentration. *Ann Intern Med*. 1989 Jun 1; 110(11):855-857 (review). PubMed 2655518

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2. Lee WH, Packer M. Prognostic importance of serum sodium concentration and its modification by converting-enzyme inhibition in patients with severe chronic heart failure. *Circulation*. 1986 Feb; 73(2):257-267. PubMed 3002660 (<http://www.ncbi.nlm.nih.gov/pubmed/3002660>)

3. Aw TC, Kiechle FL. Pseudohyponatremia. *Am J Emerg Med*. 1985 May; 3(3):236-239. PubMed 3994801 (<http://www.ncbi.nlm.nih.gov/pubmed/3994801>)

4. Epstein M, Oster JR. Disorders of hyponatremia and hypernatremia. In: Halsted JA, Halsted CH, eds. *The Laboratory in Clinical Medicine*.

Interpretation and Application. 2nd ed. Philadelphia, Pa: WB Saunders Co; 1981:289-295.

5. Votey SR, Peters AL, Hoffman JR. Disorders of water metabolism: Hyponatremia and hypernatremia. *Emerg Med Clin North Am*. 1989 Nov; 7(4):749-769 (review). PubMed 2680464

(<http://www.ncbi.nlm.nih.gov/pubmed/2680464>)

6. Kleinschmidt-DeMasters BK, Norenberg MD. Rapid correction of hyponatremia causes demyelination: Relation to central pontine myelinolysis. *Science*. 1981 Mar 6; 211(4486):1068-1070. PubMed 7466381

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7. Nelson DC, McGrew WR Jr, Hoyumpa AM Jr. Hypernatremia and lactulose therapy. *JAMA*. 1983 Mar 11; 249(10):1295-1298. PubMed 6827705 (<http://www.ncbi.nlm.nih.gov/pubmed/6827705>)

8. Hitz J, Trivin F. Sodium. In: Siest G, Galteau MM, eds. *Drug Effects on Laboratory Test Results Analytical Interferences and Pharmacological Effects*. Littleton, Mass: PSG Publishing Co;1988:391-404.